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Module 1.4 Market Structures

1. Overview of Market Structures

Feature	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of firms	Very many	Many	Few (dominated by a handful)	One
Product	Homogeneous (identical)	Differentiated	Differentiated or homogeneous	Unique (no close substitutes)
Barriers to entry	None	Low	High	Very high
Price-setting power	None (price taker)	Some	Significant (interdependent)	Price maker
Long-run profit	Normal only	Normal only	Supernormal possible	Supernormal possible
Examples	Currency markets, agricultural commodities	Restaurants, hairdressers, clothing	Supermarkets, airlines, mobile networks	Thames Water, Network Rail

2. Perfect Competition

Perfect competition is a theoretical market structure with many buyers and sellers, homogeneous products, perfect information, free entry and exit, and no individual firm able to influence the market price.

2.1 Assumptions

- **Many buyers and sellers** — each too small to influence the market price
- **Homogeneous products** — identical goods, so consumers have no preference between firms
- **Perfect information** — all buyers and sellers know all prices and product quality
- **Free entry and exit** — no barriers; firms can enter or leave the market costlessly
- **No transport costs** — goods are costless to move
- **Profit maximisation** — all firms aim to maximise profit

The firm is a **price taker**: it faces a perfectly elastic (horizontal) demand curve at the market price. It can sell any quantity at the market price but nothing above it.

2.2 Short-Run Equilibrium — Supernormal Profit

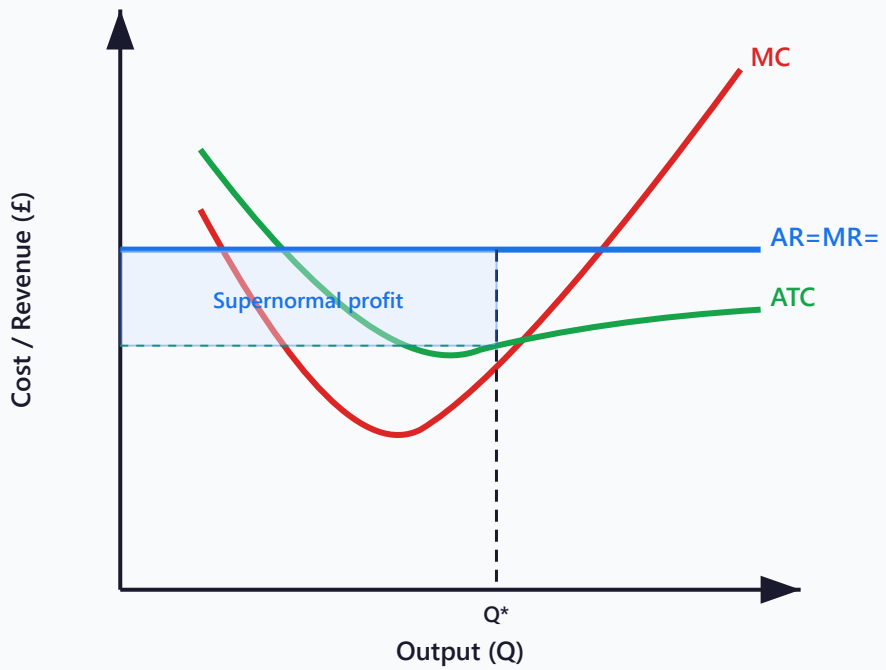

Figure 1: Perfect competition — short-run supernormal profit where P > ATC

2.3 Long-Run Equilibrium — Normal Profit Only

In the long run, supernormal profit attracts new firms (free entry). Market supply increases, pushing the market price down until firms earn **normal profit only** (AR = ATC). Conversely, if firms make losses, some exit, supply falls, and price rises back to normal profit.

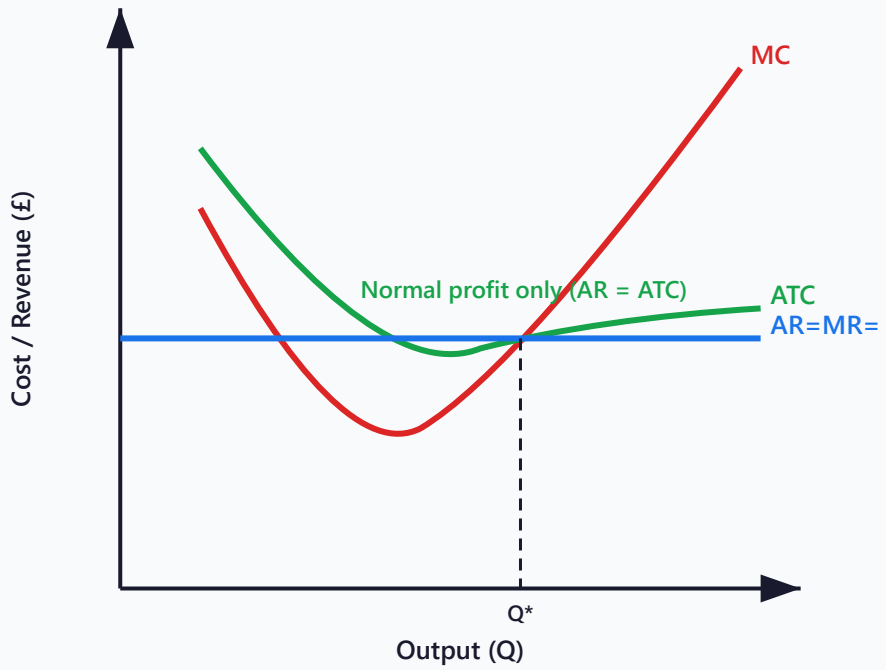

Figure 2: Perfect competition — long-run equilibrium at normal profit, P = min ATC

2.4 Efficiency in Perfect Competition

- **Allocative efficiency (P = MC):** Achieved — price reflects marginal cost, so resources go where consumers value them most.
- **Productive efficiency (P = min ATC):** Achieved in the long run — firms produce at the lowest possible average cost.
- **Dynamic efficiency:** Weak — normal profit means limited funds for R&D and innovation.

3. Monopoly

Pure monopoly: A single firm supplies the entire market. In UK competition law, a firm with 25%+ market share is considered to have monopoly power.

Barriers to entry prevent new firms from entering the market, allowing the monopolist to sustain supernormal profit in the long run.

3.1 Sources of Monopoly Power (Barriers to Entry)

Barrier	Explanation	Example
Economies of scale	Existing firm has lower costs due to large scale; new entrants cannot compete at small scale	Car manufacturing, aircraft production
Legal barriers	Patents, copyrights, licences, regulations	Pharmaceutical patents (20-year protection)
Brand loyalty	Strong brands make it hard for new entrants to attract customers	Coca-Cola, Apple
Control of resources	Owning essential raw materials or distribution channels	De Beers historically controlling diamond supply
Network effects	Product becomes more valuable as more people use it	Facebook/Meta, WhatsApp
Predatory pricing	Incumbent temporarily prices below cost to drive out new entrants	Allegations against Amazon in book market
High sunk costs	Large irrecoverable investment deters entry	Airline routes (landing slots), telecoms infrastructure

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3.2 Monopoly Profit Maximisation

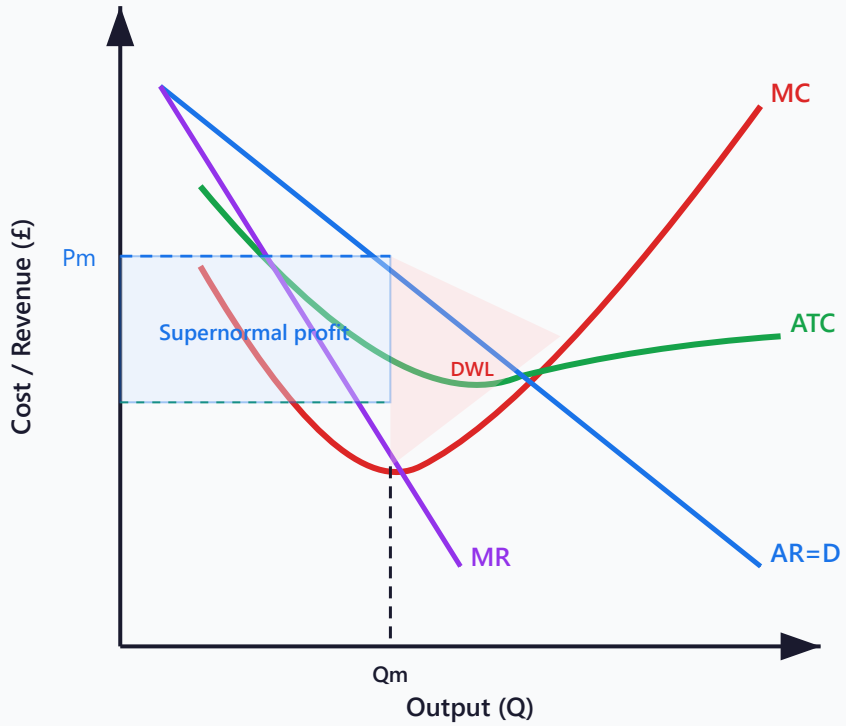

Figure 3: Monopoly — profit maximisation at MC = MR, with supernormal profit and deadweight loss (DWL)

3.3 Efficiency in Monopoly

- **Allocative efficiency:** Not achieved — P > MC, so the monopolist charges more than the marginal cost of production. Consumer surplus is reduced.
- **Productive efficiency:** Not achieved — the monopolist does not produce at the minimum of ATC (it restricts output to maximise profit).
- **X-inefficiency:** Without competitive pressure, costs may be higher than necessary (managerial slack).
- **Dynamic efficiency:** Potentially strong — supernormal profits can fund R&D and innovation (Schumpeter's argument).

3.4 Arguments For and Against Monopoly

Arguments against monopoly:

- Higher prices, lower output than under competition — welfare loss (deadweight loss)
- Allocative and productive inefficiency
- X-inefficiency due to lack of competition
- Income inequality — profits transfer from consumers to shareholders
- Rent-seeking — monopolists may spend resources lobbying to protect their position rather than innovating

Arguments for monopoly:

- Economies of scale — may have lower costs than many small firms (natural monopoly)
- Supernormal profits fund R&D and innovation (Schumpeter)
- Cross-subsidisation — profits from one product can fund development of others
- International competitiveness — large firms can compete globally
- Natural monopoly — some industries are most efficient with a single provider

Case Study: Google's Search Engine Dominance

Google holds approximately 90%+ of the global search engine market. Its dominance stems from network effects (more users attract more advertisers, funding better algorithms), massive economies of scale in data processing, brand loyalty, and the superior quality of its search results. The European Commission fined Google over 8 billion euros between 2017 and 2019 for anti-competitive practices including favouring its own shopping results and restricting Android manufacturers. Critics argue Google restricts competition and consumer choice. Supporters counter that Google's dominance reflects superior quality and that barriers are partly natural (scale requirements for an effective search engine are enormous).

4. Monopolistic Competition

Monopolistic competition is a market structure with many firms selling differentiated products, low barriers to entry and exit, and each firm having a small degree of market power.

4.1 Characteristics

- **Many firms** — each with a small market share
- **Product differentiation** — branding, quality, location, service create differences (gives each firm a small downward-sloping demand curve)
- **Low barriers to entry and exit** — easy for new firms to set up
- **Non-price competition** — advertising, loyalty schemes, quality improvements
- **Imperfect information** — consumers may not know all options available

4.2 Short Run — Supernormal Profit Possible

In the short run, a monopolistically competitive firm can earn supernormal profit by differentiating its product (similar diagram to monopoly).

4.3 Long Run — Normal Profit

Supernormal profit attracts new firms (low barriers). New entrants take market share from existing firms, shifting each firm's demand curve leftward until only **normal profit** remains (AR tangent to ATC).

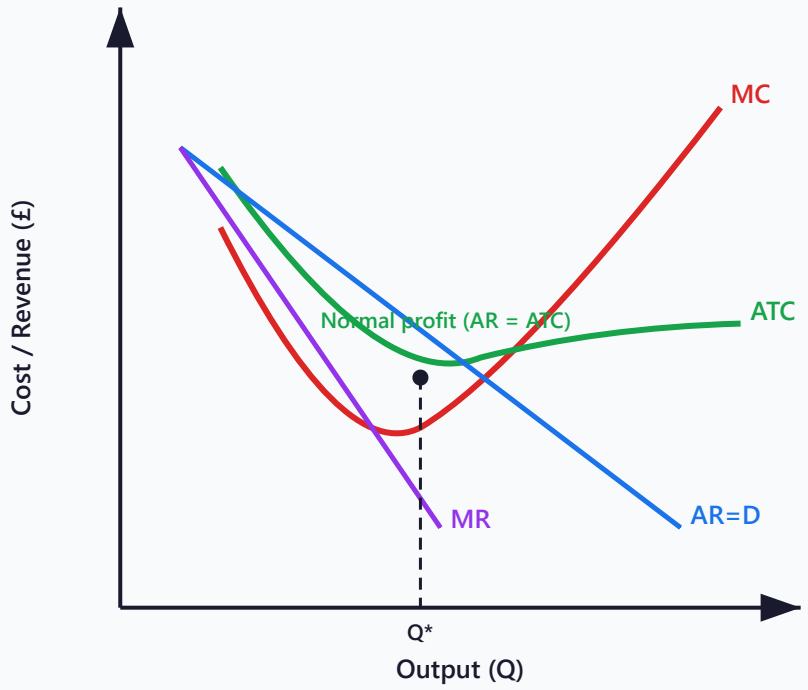

Figure 4: Monopolistic competition — long-run equilibrium at normal profit (AR tangent to ATC)

4.4 Efficiency

- **Allocative efficiency:** Not achieved — P > MC
- **Productive efficiency:** Not achieved — firm produces to the left of minimum ATC (excess capacity)
- However, consumers benefit from **product variety and choice**, which may offset the efficiency losses

5. Oligopoly

Oligopoly is a market structure dominated by a few large firms, each with significant market share. The defining feature is **interdependence** — each firm must consider the likely reactions of rivals when making pricing and output decisions.

5.1 Characteristics

- **Few dominant firms** — high concentration ratio (the top 3-5 firms control a large share of the market)
- **Interdependence** — the key distinguishing feature; each firm's decisions affect and are affected by rivals
- **High barriers to entry** — economies of scale, brand loyalty, patents, capital requirements
- **Product differentiation** — may be strong (cars, smartphones) or weak (oil, steel)
- **Non-price competition** — advertising, branding, quality, customer service (price wars are risky)
- **Price rigidity** — prices tend to be stable (explained by the kinked demand curve theory)

5.2 Concentration Ratio

The **n-firm concentration ratio** measures the combined market share of the n largest firms. A high ratio indicates oligopoly. For example, a 5-firm concentration ratio of 80% means the top 5 firms control 80% of market sales.

5.3 The Kinked Demand Curve

The kinked demand curve model (Sweezy) explains **price rigidity** in oligopoly:

- If a firm **raises** its price, rivals do not follow — the firm loses customers to competitors. Demand is **elastic** above the current price.
- If a firm **lowers** its price, rivals match the cut to protect their market share. The firm gains few extra customers. Demand is **inelastic** below the current price.
- This creates a **kink** in the demand curve at the current price and a **discontinuity in the MR curve**.
- MC can shift within the discontinuity without changing the profit-maximising price — explaining price stability.

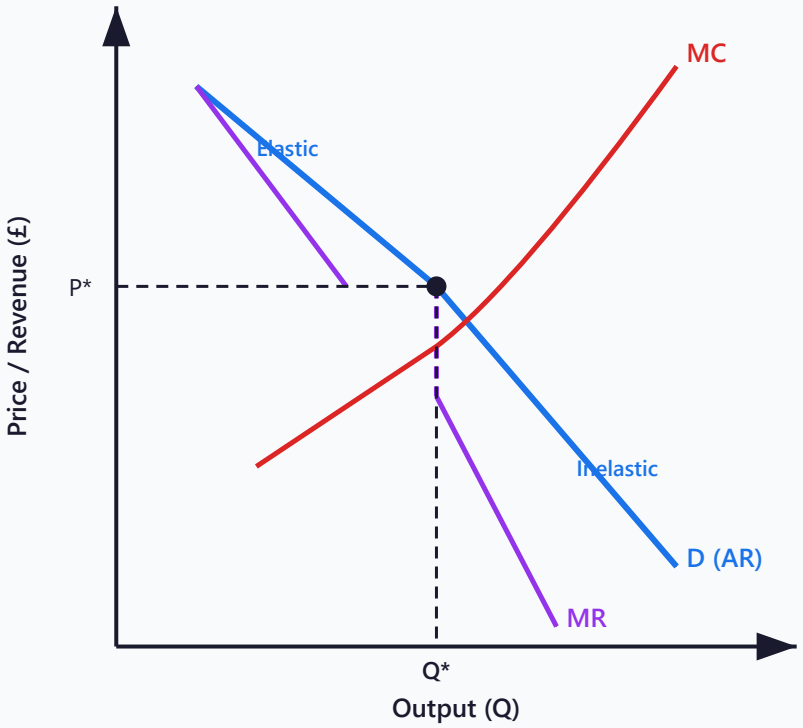

Figure 5: Kinked demand curve — price rigidity in oligopoly. MC can shift within the MR gap without changing price.

5.4 Game Theory and the Prisoner's Dilemma

Game theory analyses strategic decision-making where outcomes depend on the actions of all players. The **prisoner's dilemma** illustrates why oligopolists may fail to collude even when cooperation would benefit them all.

Consider two firms deciding whether to set high or low prices:

	Firm B: High Price	Firm B: Low Price
Firm A: High Price	Both earn £10m (best joint outcome)	A: £2m, B: £15m
Firm A: Low Price	A: £15m, B: £2m	Both earn £5m (Nash equilibrium)

The **dominant strategy** for each firm is to set a low price (regardless of what the other does). The **Nash equilibrium** (both choose low price, earning £5m each) is worse for both than if they cooperated (both high price, £10m each). This explains why collusion is tempting but unstable — there is always an incentive to cheat.

5.5 Collusion

Overt (Formal) Collusion — Cartels

Firms explicitly agree to fix prices, restrict output, or divide markets. This is **illegal** in the UK and EU under competition law.

Tacit (Informal) Collusion

Firms coordinate behaviour without explicit agreement, through price leadership (following the dominant firm's pricing) or parallel pricing. This is harder to detect and prosecute.

Factors Favouring Collusion

- Few firms (easier to coordinate and monitor)
- Similar cost structures (easier to agree on prices)
- High barriers to entry (outsiders cannot undercut)
- Homogeneous products (less incentive to compete on quality)
- Stable demand (reduces incentive to cheat)

Why Cartels Break Down

- Incentive to cheat — each firm can increase profit by secretly cutting prices
- Difficulty monitoring compliance among many firms
- Different cost structures make agreement difficult

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- New entrants may undercut the cartel price
- Recessions increase incentive to grab market share
- Legal penalties — heavy fines and criminal prosecution deter overt collusion

Case Study: UK Supermarket Oligopoly

The UK grocery market is a classic oligopoly. Four firms (Tesco, Sainsbury's, Asda, Morrisons) control approximately 65% of the market, with discounters Aldi and Lidl gaining share. Interdependence is evident — when Tesco launched its "Price Promise" campaign, rivals immediately matched with their own guarantees. Competition is largely through non-price methods: loyalty cards (Tesco Clubcard, Nectar), home delivery, premium own-brand ranges, and store layouts. The Competition and Markets Authority (CMA) closely monitors the sector for anti-competitive behaviour. The attempted Sainsbury's-Asda merger in 2019 was blocked by the CMA over concerns it would reduce competition and raise prices.

6. Contestable Markets

A contestable market (Baumol) is one where there is freedom of entry and exit, and where the threat of potential competition constrains the behaviour of existing firms. Sunk costs are zero or very low. Even a monopolist in a contestable market may behave competitively.

6.1 Key Features

- **Low or zero sunk costs** — firms can enter and exit without losing investment (sunk costs are the key barrier)
- **Hit-and-run entry** — if incumbents charge high prices, new firms can enter, earn profit, and exit before the incumbent responds
- **Threat of entry** — the mere possibility of entry disciplines existing firms to keep prices low and efficiency high

6.2 Implications

In a perfectly contestable market, even a monopolist would set price at average cost (normal profit only) to deter entry. The number of firms in the market matters less than the **threat of competition**.

6.3 Factors Affecting Contestability

- **Sunk costs:** High sunk costs reduce contestability (e.g. rail infrastructure vs a food truck)
- **Regulation:** Deregulation increases contestability (e.g. airline deregulation in the 1980s)
- **Technology:** Internet reduces barriers (e.g. online retailing contests brick-and-mortar shops)
- **Brand loyalty:** Strong brands reduce contestability

Case Study: Airline Industry and Contestability

The airline industry became significantly more contestable after deregulation in the 1980s-90s. Budget airlines like Ryanair and easyJet entered markets previously dominated by national carriers (British Airways, Lufthansa). Sunk costs are relatively low — aircraft can be leased and routes changed quickly. However, contestability is limited by landing slot scarcity at major airports (Heathrow slots are extremely valuable and rarely available), loyalty programmes (frequent flyer schemes lock in customers), and economies of scale in fleet management. This shows that real markets are on a spectrum of contestability rather than perfectly contestable.

7. Efficiency Comparisons

Allocative efficiency: P = MC — resources allocated to their highest-valued use.
Productive efficiency: Production at the minimum of ATC — lowest possible cost per unit.
Dynamic efficiency: Improvement in products and processes over time through innovation and R&D.
X-efficiency: Producing at the lowest cost given existing technology (no managerial slack).

Efficiency Type	Perfect Competition	Monopoly	Oligopoly
Allocative	Yes (P = MC)	No (P > MC)	No (P > MC)

Productive	Yes (long run)	No (not at min ATC)	Possibly (competitive pressure)
Dynamic	Weak (no supernormal profit for R&D)	Potentially strong (profits fund innovation)	Often strong (profits + competitive pressure)
X-efficiency	Strong (competitive pressure)	Weak (no pressure to minimise costs)	Moderate

Exam tip: The Schumpeter argument is crucial for evaluation — monopoly profits can fund innovation that benefits consumers in the long run (dynamic efficiency). This is the strongest counter-argument to the standard case against monopoly. Always mention it in essays on monopoly.

8. The Labour Market

8.1 Demand for Labour

Demand for labour is a **derived demand** — firms demand workers not for their own sake but for the output they produce. The demand for labour depends on the demand for the product the labour produces.

Marginal Revenue Product (MRP)

$$\text{MRP} = \text{MPP} \times \text{MR (Marginal Physical Product} \times \text{Marginal Revenue)}$$

A profit-maximising firm hires workers up to the point where **MRP = Wage rate (MCL)**. The MRP curve is the firm's demand curve for labour.

Factors Shifting Labour Demand

- Changes in demand for the product (derived demand)
- Changes in labour productivity (technology, training)

- Changes in the price of substitute factors (e.g. automation becoming cheaper)
- Changes in the price of complementary factors

8.2 Supply of Labour

The supply of labour to a particular occupation depends on:

- **Wage rate** (higher wages attract more workers)
- **Non-monetary factors** (working conditions, job satisfaction, location, prestige)
- **Qualifications and training requirements** (barriers to entry)
- **Population size and demographics**
- **Migration** (international movement of workers)
- **Trade unions** (may restrict labour supply to push up wages)

8.3 Wage Determination

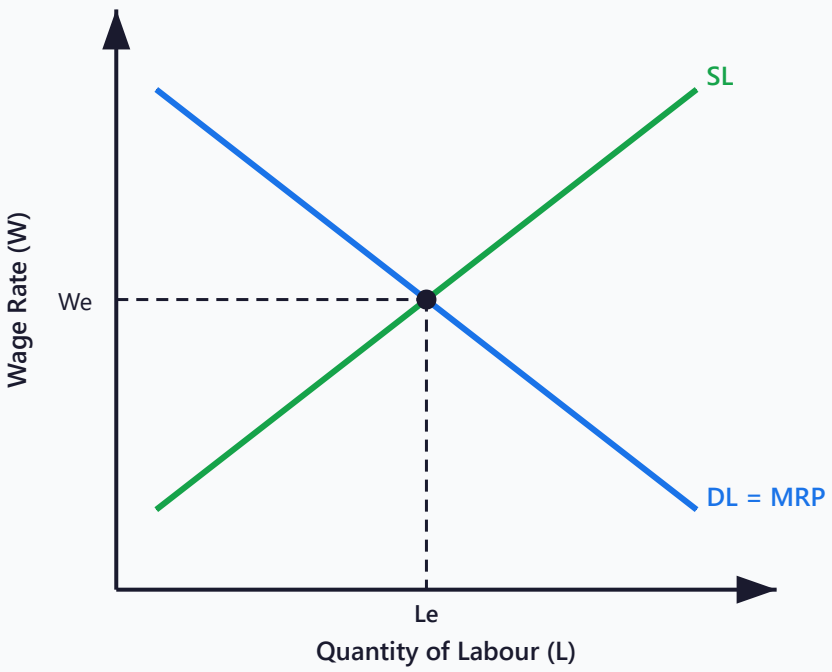

Figure 6: Wage determination in a competitive labour market — equilibrium at SL = DL

8.4 Wage Differentials

Wages differ across occupations and workers due to:

- **Compensating wage differentials:** Higher wages for dangerous, unpleasant, or unsocial hours (e.g. oil rig workers, night shift workers).
- **Human capital:** Education, training, experience, and skills increase productivity and command higher wages.
- **Barriers to entry:** Professional qualifications (doctors, lawyers) restrict supply and raise wages.
- **Trade union power:** Unions can negotiate wages above the market-clearing level.
- **Monopsony power:** A dominant employer (like the NHS for nurses) can suppress wages below the competitive level.
- **Discrimination:** Gender, ethnic, or age discrimination can lead to unequal pay for equal work.

8.5 Monopsony in the Labour Market

> **Monopsony** exists when there is a single (or dominant) buyer of labour. The monopsonist has the power to set wages below the competitive equilibrium because workers have limited alternative employers.

Examples: NHS as a major employer of nurses, a single large factory in a small town, the army as the sole employer of soldiers.

A monopsonist faces an upward-sloping supply curve (must raise wages to attract more workers). The marginal cost of labour (MCL) lies above the supply curve because raising the wage for one extra worker also raises it for all existing workers.

> **Evaluation — Market Structures and Welfare**
>
> - No market structure is unambiguously "best" — each involves trade-offs between static efficiency, dynamic efficiency, and consumer choice.
> - Perfect competition maximises static efficiency but may lack innovation incentives.
> - Monopoly may deliver dynamic efficiency through R&D but at the cost of higher prices and restricted output.
> - Oligopoly can combine competitive pressure with innovation capacity, but may also lead to collusion and consumer harm.

- Contestability may matter more than market structure — the threat of competition can discipline any market.
- Government policy (competition law, regulation, deregulation) aims to capture the benefits of each structure while minimising the costs.

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